Cafe revenue is driven by price, not popularity, and holds steady all year

Executive summary | Exploratory Data Analysis of 10,000 cafe transactions | DecodeLabs Data Analytics, Project 2 | Python (pandas, seaborn, scipy)

1. Problem statement

A cafe owner wants to know which items drive revenue, whether in-store and takeaway customers behave differently, whether sales follow a seasonal or weekday pattern, and whether any unusual values need attention, before investing in dashboards or forecasts.

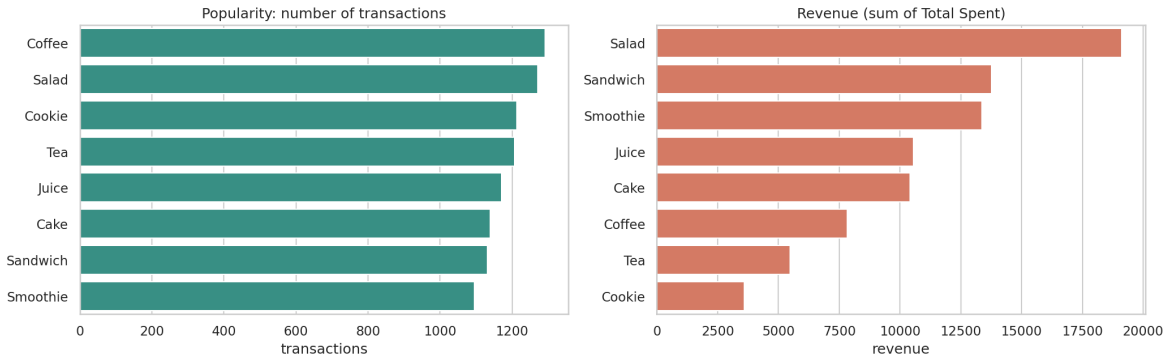
2. Methodology

The analysis uses the audited dataset from Project 1 (10,000 transactions, zero duplicate IDs, all dates ISO-formatted). Gaps that could not be reliably filled were left in place, so each question uses only the rows it can trust: the 460 undated rows were excluded from trend analysis, and records with an unknown item, location or payment method were excluded from the matching comparison. Steps: descriptive statistics (mean, median, count, skewness), distribution checks, outlier detection with both the IQR and Z-score methods, category comparisons with Mann-Whitney and Kruskal-Wallis tests (chosen because order values are skewed), month and weekday trends normalised per day, and correlation analysis.

3. Key findings

Metric	Result
Transactions / total revenue	10,000 / $89,274.50
Average vs median order	$8.93 vs $8.00 (right-skewed, skewness 0.82)
Top item by revenue	Salad: $19,125, 22.7% of known-item revenue
Most frequently sold item	Coffee: 1,291 transactions (sixth by revenue)
Lowest earner	Cookie: $3,604
Unit price vs item revenue (correlation)	0.99 (transactions vs revenue: -0.11)
Outliers in Total Spent	IQR flags 269; Z-score flags 0. 268 are Salad x 5 (valid maximum orders)
In-store vs takeaway average order	$9.04 vs $8.81 (p = 0.14, no meaningful difference)
Payment methods	No meaningful difference in order value (p = 0.77)
Monthly revenue per day variation	2.7% across months; weekdays differ by at most 1.06x

- **Price decides the revenue ranking.** Every item sells at a similar rate (1,096 to 1,291 transactions), so the high-priced items (Salad, Sandwich, Smoothie) lead on revenue while low-priced Cookie and Tea trail despite similar popularity.
- **The big orders are real.** The IQR rule flags 269 high-value orders only because the pooled order value is skewed. Nearly all are five Salads, the largest possible order, so they are signal and were kept. One flagged row is an incomplete record carried over from the cleaning stage and is noted for review.
- **No strong seasonality, weekday effect, or customer-type split.** Monthly revenue per day moves within a narrow band (peak June, low July, a small gap), and in-store and takeaway orders are statistically indistinguishable.



4. Recommendations

- **Test, don't assume, a price-led strategy.** Feature higher-priced items in menus and promotions, and test bundles or add-ons for Cookie and Tea (they sell as often as the leaders but earn far less). This analysis shows association only, so confirm with a controlled promotion test.
- **Add cost data before choosing winners.** Revenue is not profit. Margin data per item would show whether high-revenue items are also the most profitable.
- **Avoid location- or payment-specific tactics on this evidence.** Order value does not differ meaningfully by location or payment method, so the data does not support separate strategies.
- **Improve data capture at the till.** Location (40% unknown), payment method (32% unknown), item (480 transactions unknown) and date (460 missing) are the weakest fields. Capturing them reliably is the biggest gain for the next analysis.

5. Limitations

This is a practice dataset for a fictional cafe with near-uniform patterns, so the findings demonstrate a method more than a real business outcome. Conclusions about location rest on the 60% of rows with a known location and are indicative only.